

Finite-T Dynamic-Panel Bias and Estimator Choice

IVB-paper replication package

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1 Pre-specified design

The report reads versioned outputs and does not run simulations. The principal grid contains $N \in \{50, 100, 250\}$, $T \in \{8, 10, 15, 20, 30, 50\}$, and $\rho_Y \in \{0.2, 0.5, 0.8\}$. The stress grid fixes $N = 100$ and $\rho_Y = 0.5$, varies $T \in \{10, 20, 30\}$ and $\rho_D \in \{0.2, 0.5, 0.8\}$, and uses the inherited dual-role unit-effect scales $\sigma_{\alpha Z} \in \{0.5, 2.0\}$.

Table 1: Pre-specified principal and stress grids.

Design	Scenarios	N	T	rho_Y	rho_D	sigma_alpha_Z
principal	54	50, 100, 250	8, 10, 15, 20, 30, 50	0.2, 0.5, 0.8	0.5	0.5
stress	18	100	10, 20, 30	0.5	0.2, 0.5, 0.8	0.5, 2

2 Phase map

3 Bias, RMSE, and coverage

Table 2: Principal-grid bias, RMSE, coverage, and Monte Carlo precision summaries across outcome-persistence values.

Estimator	N	T	Max bias	Max RMSE	Min coverage	Max bias MCSE
Arellano-Bond sensitivity	50	8	0.114	0.531	0.976	0.023
Arellano-Bond sensitivity	50	10	0.148	0.515	0.976	0.022
Arellano-Bond sensitivity	50	15	0.069	0.457	0.944	0.020
Arellano-Bond sensitivity	50	20	0.046	0.296	0.954	0.013
Arellano-Bond sensitivity	50	30	0.028	0.254	0.946	0.011
Arellano-Bond sensitivity	50	50	0.010	0.124	0.944	0.006
Arellano-Bond sensitivity	100	8	0.024	0.556	0.976	0.025
Arellano-Bond sensitivity	100	10	0.086	0.472	0.958	0.021
Arellano-Bond sensitivity	100	15	0.042	0.287	0.956	0.013
Arellano-Bond sensitivity	100	20	0.029	0.212	0.932	0.009
Arellano-Bond sensitivity	100	30	0.007	0.120	0.936	0.005
Arellano-Bond sensitivity	100	50	0.005	0.074	0.944	0.003
Arellano-Bond sensitivity	250	8	0.054	0.361	0.968	0.016
Arellano-Bond sensitivity	250	10	0.042	0.318	0.952	0.014
Arellano-Bond sensitivity	250	15	0.009	0.192	0.942	0.009
Arellano-Bond sensitivity	250	20	0.011	0.122	0.956	0.005
Arellano-Bond sensitivity	250	30	0.006	0.071	0.926	0.003

Estimator	N	T	Max bias	Max RMSE	Min coverage	Max bias MCSE
Arellano-Bond sensitivity	250	50	0.002	0.041	0.934	0.002
FE-ADL within	50	8	0.015	0.058	0.922	0.003
FE-ADL within	50	10	0.018	0.049	0.936	0.002
FE-ADL within	50	15	0.014	0.039	0.928	0.002
FE-ADL within	50	20	0.011	0.032	0.936	0.001
FE-ADL within	50	30	0.010	0.027	0.922	0.001
FE-ADL within	50	50	0.006	0.020	0.920	0.001
FE-ADL within	100	8	0.019	0.044	0.898	0.002
FE-ADL within	100	10	0.016	0.037	0.916	0.001
FE-ADL within	100	15	0.014	0.028	0.920	0.001
FE-ADL within	100	20	0.015	0.027	0.888	0.001
FE-ADL within	100	30	0.010	0.020	0.906	0.001
FE-ADL within	100	50	0.007	0.015	0.916	0.001
FE-ADL within	250	8	0.018	0.030	0.878	0.001
FE-ADL within	250	10	0.019	0.028	0.858	0.001
FE-ADL within	250	15	0.015	0.022	0.846	0.001
FE-ADL within	250	20	0.013	0.019	0.858	0.001
FE-ADL within	250	30	0.009	0.015	0.834	0.001
FE-ADL within	250	50	0.006	0.011	0.868	0.000
split-panel jackknife FE-ADL	50	8	0.042	0.080	0.896	0.003
split-panel jackknife FE-ADL	50	10	0.031	0.062	0.898	0.002
split-panel jackknife FE-ADL	50	15	0.016	0.046	0.914	0.002
split-panel jackknife FE-ADL	50	20	0.013	0.036	0.926	0.001
split-panel jackknife FE-ADL	50	30	0.010	0.027	0.926	0.001
split-panel jackknife FE-ADL	50	50	0.005	0.020	0.920	0.001
split-panel jackknife FE-ADL	100	8	0.045	0.064	0.820	0.002
split-panel jackknife FE-ADL	100	10	0.033	0.051	0.840	0.002
split-panel jackknife FE-ADL	100	15	0.018	0.034	0.900	0.001
split-panel jackknife FE-ADL	100	20	0.014	0.027	0.912	0.001
split-panel jackknife FE-ADL	100	30	0.009	0.021	0.912	0.001
split-panel jackknife FE-ADL	100	50	0.005	0.014	0.932	0.001
split-panel jackknife FE-ADL	250	8	0.045	0.054	0.700	0.001
split-panel jackknife FE-ADL	250	10	0.032	0.040	0.722	0.001
split-panel jackknife FE-ADL	250	15	0.018	0.026	0.816	0.001
split-panel jackknife FE-ADL	250	20	0.014	0.020	0.836	0.001

Estimator	N	T	Max bias	Max RMSE	Min coverage	Max bias MCSE
split-panel jackknife FE-ADL	250	30	0.008	0.014	0.900	0.001
split-panel jackknife FE-ADL	250	50	0.005	0.010	0.908	0.000

4 Arellano-Bond sensitivity diagnostics

Arellano-Bond is a sensitivity benchmark under its own moment conditions. It is not labeled as bias-corrected fixed effects. Instruments are collapsed and limited to lags 2–3; the DGP contains no common time shocks, so this benchmark uses individual effects in differences and does not add a growing set of time-dummy instruments.

Table 3: Arellano-Bond sensitivity-benchmark failure and diagnostic rejection rates.

Design	N	T	Failure	Instruments	AR(2) < .05	Hansen < .05
principal	50	8	0	6	0.044	0.052
principal	50	10	0	6	0.046	0.054
principal	50	15	0	6	0.048	0.058
principal	50	20	0	6	0.050	0.056
principal	50	30	0	6	0.060	0.066
principal	50	50	0	6	0.056	0.066
principal	100	8	0	6	0.044	0.046
principal	100	10	0	6	0.046	0.058
principal	100	15	0	6	0.048	0.050
principal	100	20	0	6	0.052	0.062
principal	100	30	0	6	0.054	0.058
principal	100	50	0	6	0.052	0.054
principal	250	8	0	6	0.050	0.046
principal	250	10	0	6	0.056	0.062
principal	250	15	0	6	0.052	0.072
principal	250	20	0	6	0.068	0.060
principal	250	30	0	6	0.050	0.062
principal	250	50	0	6	0.066	0.066
stress	100	10	0	6	0.058	0.050
stress	100	20	0	6	0.052	0.068
stress	100	30	0	6	0.066	0.064

5 Monte Carlo stability and failures

Table 4: Monte Carlo precision, batch-stability, and estimator-failure checks.

Estimator	Cells	Stable cells	Max failure	Max bias MCSE
Arellano-Bond sensitivity	72	50	0	0.025
FE-ADL within	72	56	0	0.003
split-panel jackknife FE-ADL	72	55	0	0.003

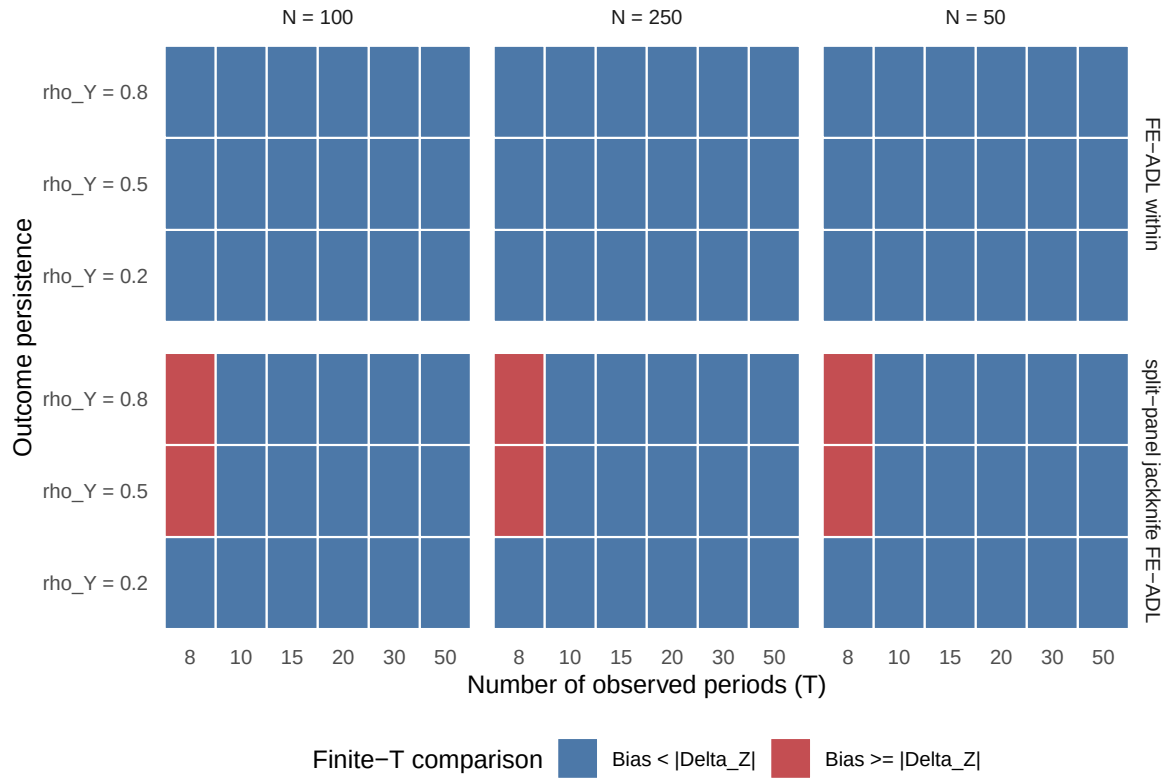


Figure 1: Phase map for whether the absolute Monte Carlo estimation bias is at least as large as the absolute mean specification shift caused by adding lagged Z . Arellano-Bond is excluded from this main comparison and retained as a sensitivity benchmark.